

Comprehensive Glossary of CCS and Carbon Market Terms

[Carbon Capture and Storage](#) | [Carbon Markets](#) | [Agricultural CDR](#) | [Climate Policy](#)

80+ Terms | Alphabetically Sorted | Cross-Referenced | Module Index

This glossary defines key terms used throughout the Climate Change: Carbon Capture and Storage course. Terms are organised alphabetically. Cross-references are shown in italics. Where a term has a specific technical definition in a carbon methodology (e.g. VM0042, Gold Standard), that definition is used. This glossary covers CCS technology, geological storage, carbon markets, agricultural CDR, project development, MRV, and climate policy.

A

Additionality

A core criterion for carbon credit validity. Emission reductions are 'additional' if they would not have occurred in the absence of the carbon project and its associated carbon finance. A project fails the additionality test if the activity would have happened anyway — for example, if a renewable energy project would have been built for purely economic reasons without carbon revenue.

Example: A no-till farming practice adopted by only 2% of farmers in a region is likely additional, because it is not yet common practice and requires behaviour change incentivised by carbon payments.

See also: Baseline, Common Practice Test

Article 6 (Paris Agreement)

The provisions of the Paris Agreement that govern international carbon market mechanisms. Article 6.2 enables bilateral country-to-country trading of Internationally Transferred Mitigation Outcomes (ITMOs). Article 6.4 establishes a new UN-supervised carbon crediting mechanism (successor to the CDM). Article 6.8 covers non-market cooperative approaches.

Example: India selling ITMOs to Japan under Article 6.2, counted toward Japan's NDC.

See also: ITMO, NDC, CDM

Amine Scrubbing

The most commercially mature post-combustion CO₂ capture technology. Flue gas is passed through a liquid amine solvent (most commonly monoethanolamine, MEA) that selectively absorbs CO₂. The CO₂-rich solvent is then heated in a regenerator to release a concentrated CO₂ stream, regenerating the solvent for reuse. Energy penalty: 20–30% of plant output.

Example: Used at Boundary Dam (Shell Cansolv process) and Norcem Brevik (Aker Carbon Capture process).

See also: Post-Combustion Capture, Solvent Regeneration

Agroforestry

The intentional integration of trees and shrubs into crop and animal farming systems. Combines agricultural and forestry practices for ecological, economic, and social benefits. As a carbon project activity, agroforestry sequesters carbon in above-ground woody biomass and in soil organic matter from root turnover and leaf litter.

Example: Planting Moringa trees at 100/hectare between rice rows in a Bihar farming system.

See also: Soil Carbon, CDR, Regenerative Agriculture

Aquifer (Deep Saline)

A deep underground layer of porous rock saturated with saline water — too salty for drinking or agriculture. Deep saline aquifers are the most abundant geological formation suitable for CO₂ storage, with a global estimated capacity of 100,000+ GtCO₂.

Example: The Utsira sandstone (Sleipner project) and the Johansen formation (Northern Lights) are both deep saline aquifers.

See also: Geological Storage, Trapping Mechanisms

B

Baseline Emissions

The emissions that would have occurred in the absence of the carbon project — the 'business as usual' scenario. Accurately defining the baseline is critical to quantifying the project's additionality: credits are issued only for reductions below the baseline. The baseline must be based on evidence, not assumption, and must be conservative (not overstating what would have happened).

Example: A village cluster where 80% of farmers open-burn crop residue each year: this is the baseline for a biochar project.

See also: Additionality, Carbon Accounting

BECCS

Bioenergy with Carbon Capture and Storage. Burning sustainably-sourced biomass for energy (electricity or heat), capturing the resulting CO₂ with CCS technology, and storing it geologically. Because the biomass absorbed CO₂ while growing, and that CO₂ is now permanently stored underground, BECCS achieves net-negative emissions. Featured in nearly all IPCC 1.5C scenarios as a critical CDR technology.

Example: Drax Power Station in the UK — the world's largest biomass power plant — is developing BECCS to achieve net-negative emissions.

See also: CDR, CCS, Carbon Budget

Benefit Sharing

The mechanism by which carbon project revenues are distributed among participants — typically farmers or community members — and implementing organisations. A well-designed benefit sharing framework ensures that the primary contributors to carbon sequestration (the farmers) receive a meaningful share of the revenue (typically 50–70%), building long-term participation and project credibility.

Example: An FPO distributes 65% of carbon credit revenue directly to enrolled farmers, 10% to a community development fund, and 25% to operational costs.

See also: Carbon Credits, FPO, Additionality

Biochar

A carbon-rich, porous solid material produced by heating organic biomass (crop residue, wood waste, manure) in the absence of oxygen — a process called pyrolysis. Biochar is highly stable in soil, with a half-life of 100–10,000+ years. Applied to agricultural soil, it sequesters carbon, improves water and nutrient retention, and increases crop yields. Under the VM0042 methodology, 1 tonne of biochar sequesters approximately 2.4–3.0 tonnes of CO₂ equivalent.

Example: Rice straw pyrolysed at 500C in a TLUD kiln yields approximately 25–30% biochar by dry weight.

See also: Pyrolysis, TLUD, VM0042, Carbon Sequestration

Blue Hydrogen

Hydrogen produced from natural gas via Steam Methane Reforming (SMR) combined with carbon capture and storage, capturing 85–95% of the CO₂ produced. Distinguished from 'green hydrogen' (produced from renewable electricity via electrolysis) and 'grey hydrogen' (SMR without CCS). Blue hydrogen offers a lower-cost near-term pathway to large-scale hydrogen supply, though concerns about methane leakage from the gas supply chain affect its overall lifecycle emissions.

Example: Shell's Quest project produces blue hydrogen at Scotford, Alberta, capturing ~1 Mt CO₂/year.

See also: Hydrogen, SMR, Pre-Combustion Capture, Grey Hydrogen

Buffer Pool

A reserve of unissued carbon credits held back from a project's total verified credits to cover the risk of future reversals. Under Verra VCS, typically 10–30% of credits are contributed to a buffer pool. If the project suffers a reversal event (fire, drought, land use change), credits are taken from the buffer pool rather than requiring the buyer to return credits.

Example: A biochar project with 2,730 annual gross credits contributes 409 credits (15%) to the buffer pool, issuing 2,321 net credits.

See also: Carbon Credits, Reversal, Permanence

C

Carbon Accounting

The systematic measurement, reporting, and verification of greenhouse gas emissions and removals. In carbon project context, it includes defining project boundaries, establishing baselines, quantifying reductions or removals using an approved methodology, applying uncertainty deductions, and producing a verified figure for credit issuance.

Example: Applying VM0042 equations to calculate that 7 kg of biochar per batch sequesters approximately 18–21 kg CO₂e.

See also: MRV, Methodology, Additionality

Carbon Border Adjustment Mechanism (CBAM)

A European Union trade policy measure that applies a carbon cost to imports of certain carbon-intensive goods (cement, steel, aluminium, fertilisers, electricity, and hydrogen from 2026) from countries without equivalent carbon pricing. CBAM is designed to prevent 'carbon leakage' — where EU producers are undercut by cheaper imports from countries with weaker climate rules. For India, CBAM is a significant driver of industrial decarbonisation pressure.

Example: Indian cement exported to the EU will face CBAM charges based on its embodied carbon content from 2026 onwards.

See also: Carbon Pricing, EU ETS, CCS

Carbon Budget

The total cumulative amount of CO₂ (or CO₂ equivalent) that can be emitted into the atmosphere while limiting global average temperature rise to a specified level. Carbon budgets are defined probabilistically. As of 2024, the remaining 1.5C carbon budget (50% probability) is approximately 250 GtCO₂ — at current emissions of ~40 Gt/year, this will be exhausted in roughly 6 years.

Example: If the remaining 1.5C budget is 250 GtCO₂ and global emissions are 40 Gt/year, the budget is exhausted in 6.25 years.

See also: Net-Zero, Carbon Dioxide Removal, IPCC AR6

Carbon Credit / Carbon Offset

A tradeable financial instrument representing one tonne of CO₂ equivalent that has been reduced, avoided, or removed from the atmosphere, and verified by an accredited third party. In compliance markets, these are called allowances or permits; in voluntary markets they are called carbon credits or offsets (Verified Carbon Units, Gold Standard VERs, etc.). A buyer who 'retires' a credit can claim the corresponding emission reduction.

Example: An NGO retires 500 VCUs from a Gold Standard soil carbon project to achieve a carbon-neutral event.

See also: Carbon Market, VCS, Gold Standard, Retirement

Carbon Dioxide Removal (CDR)

Activities that actively remove CO₂ from the atmosphere and durably store it — either in geological formations, in soil, in biomass, or in products. CDR is distinct from emission avoidance (which prevents new emissions) and is essential for compensating residual emissions in net-zero scenarios. CDR pathways include: afforestation/reforestation, soil carbon enhancement, biochar, BECCS, direct air capture (DAC), and enhanced weathering.

Example: The IPCC AR6 suggests 6–16 GtCO₂/year of CDR may be needed globally by 2050 in 1.5C scenarios.

See also: Biochar, BECCS, DAC, Net-Zero

Carbon Leakage

The increase in emissions in one region resulting from climate policy in another region — typically when production shifts to areas with less stringent regulations. For example, if EU climate regulations increase cement production costs, some cement production may shift to India or China, where emissions per tonne may be higher, partially offsetting the EU's mitigation effort.

Example: CBAM is explicitly designed to prevent carbon leakage in cement, steel, and aluminium trade.

See also: CBAM, EU ETS, Additionality

Carbon Market

A trading system in which carbon emissions are reduced by trading emission allowances or carbon credits. Compliance markets are mandatory, regulated by government (e.g. EU ETS, California Cap-and-Trade). Voluntary Carbon Markets (VCMs) are driven by corporate sustainability commitments and operate under private standards (Verra, Gold Standard). Both markets have a price for carbon — the amount buyers pay per tonne of CO₂ equivalent.

Example: The EU ETS carbon price reached over 100/tCO₂ in 2022, making CCS economically viable for many European industrial applications.

See also: Carbon Credit, EU ETS, Voluntary Carbon Market

Carbon Pricing

A policy mechanism that puts an explicit price on greenhouse gas emissions, either through a carbon tax (fixed price per tonne) or through an emissions trading system (cap-and-trade, where price is market-determined). Carbon pricing is widely considered the most economically efficient policy for driving emissions reductions across the economy.

Example: Norway's offshore CO₂ tax of approximately \$50–80/tonne was the direct cause of the Sleipner CCS project being economically viable.

See also: EU ETS, Carbon Tax, CCS

CCS (Carbon Capture and Storage)

A set of technologies that capture CO₂ from large point sources (power plants, industrial facilities) or from the atmosphere (DAC), transport the captured CO₂, and store it permanently in geological formations. CCS is complementary to renewable energy — it addresses sectors where renewables cannot eliminate CO₂ emissions: cement calcination, steel production, chemical processes, and the residual fossil fuel use required during the energy transition.

Example: The Sleipner project has stored over 25 million tonnes of CO₂ in the North Sea Utsira aquifer since 1996.

See also: Capture, Transport, Storage, Hard-to-Abate

CDM (Clean Development Mechanism)

A mechanism under the Kyoto Protocol that allowed developed countries (Annex I) to fund emission reduction projects in developing countries, earning Certified Emission Reductions (CERs) toward their compliance obligations. The CDM ran from 2006 to approximately 2020 and registered thousands of projects globally. It is being replaced by Article 6.4 of the Paris Agreement.

Example: India was one of the largest CDM host countries, registering over 1,500 CDM projects.

See also: Article 6, Carbon Credits, NDC

Clinker

The intermediate product of cement production — calcium silicate nodules produced by heating limestone and clay to approximately 1,450C in a rotary kiln. Grinding clinker with gypsum produces Portland cement. The production of clinker accounts for the majority of cement's CO₂ emissions, both from fuel combustion and from limestone calcination.

Example: Each tonne of clinker produces approximately 0.7 tonnes of CO₂ from calcination alone.

See also: Cement, Calcination, CCS

Corresponding Adjustment

The accounting mechanism under Article 6 of the Paris Agreement that prevents double-counting of emission reductions. When a carbon credit (ITMO) is transferred from one country to another, the selling country must 'correspondingly adjust' its NDC accounts — adding the transferred reduction back to its accounting, so it cannot also claim it domestically. Only the buying country claims it.

Example: India sells 100,000 ITMOs to Japan. India's NDC progress decreases by 100,000 tonnes; Japan's increases.

See also: ITMO, NDC, Article 6

Crop Residue

The organic material remaining in a field after crop harvest — including straw, stalks, leaves, and roots. In India, approximately 65 million tonnes of crop residue (primarily rice straw in Punjab, Haryana, and UP) is burned annually in fields, releasing CO₂, methane, and particulates. As a biochar feedstock, this same crop residue becomes a carbon sequestration resource rather than a pollution source.

Example: Rice straw from one hectare of paddy (approximately 5–7 tonnes dry weight) can produce 1.25–2.1 tonnes of biochar.

See also: Biochar, Pyrolysis, Open Burning

D

DAC (Direct Air Capture)

A technology that removes CO₂ directly from ambient air (at a concentration of approximately 420 ppm) using solid sorbents (temperature or vacuum swing systems) or liquid solvents (typically potassium hydroxide). DAC can be located anywhere, is not dependent on proximity to emission sources, and produces a highly pure CO₂ stream suitable for geological storage or industrial use. Current cost: \$250–600/tCO₂; projected to fall to \$100–200/tCO₂ by 2040 with scale-up.

Example: Climeworks' Mammoth plant in Iceland captures 36,000 tonnes of CO₂/year from ambient air, injecting it into basalt rock for permanent mineralisation.

See also: CDR, Carbon Sequestration, Geological Storage

DMRV (Digital MRV)

The application of digital technologies — including satellite remote sensing, drone surveys, IoT sensors, mobile data capture applications, machine learning, and blockchain registries — to Monitoring, Reporting, and Verification of carbon projects. DMRV reduces the cost of ground-based monitoring by 40–70%, improves data quality and traceability, and enables real-time or near-real-time monitoring at scale.

Example: A smallholder biochar project uses KoboToolbox for batch data capture, Sentinel-2 satellite imagery for field boundary verification, and Toucan Protocol for blockchain-based credit issuance.

See also: MRV, Blockchain, Satellite Monitoring

DRI (Direct Reduced Iron)

A method of producing iron by reducing iron ore (removing oxygen) using a reducing gas — typically natural gas or, increasingly, hydrogen — without melting the ore. DRI production has a significantly lower CO₂ intensity than traditional blast furnace iron production when using natural gas, and near-zero CO₂ when using green or blue hydrogen. DRI is melted in electric arc furnaces (EAF) to produce steel.

Example: HYBRIT in Sweden produces hydrogen-based DRI using 100% green hydrogen, with near-zero CO₂ steel output.

See also: Steel, Hydrogen, CCS, Hard-to-Abate

E

Emission Reduction

A decrease in greenhouse gas emissions relative to a baseline scenario. Emission reductions may be 'avoided' (preventing an emission that would otherwise have occurred) or 'sequestered' (removing CO₂ from the atmosphere and storing it). In carbon markets, only verified, additional emission reductions generate carbon credits.

Example: A biochar project prevents the open burning of crop residue (avoiding CH₄ and N₂O emissions) while also sequestering carbon in the biochar applied to soil.

See also: Carbon Credits, Additionality, CDR

Enhanced Oil Recovery (EOR)

A technique for extracting additional oil from partially depleted reservoirs by injecting fluids — including CO₂ — to maintain reservoir pressure and mobilise remaining oil. CO₂-EOR simultaneously stores CO₂ underground and generates additional oil production revenue. However, the climate accounting is complex: while the CO₂ is stored, the additional oil produced will be burned, releasing CO₂.

Example: The Boundary Dam project sells captured CO₂ to the Weyburn oilfield for CO₂-EOR, 66 km away.

See also: CCS, Geological Storage, Carbon Accounting

EU ETS (European Union Emissions Trading System)

The world's first and largest mandatory carbon market, covering approximately 10,500 power plants and industrial facilities in 30 European countries. The EU ETS operates on a cap-and-trade basis: a declining cap on total emissions, with allowances auctioned or allocated to covered entities who must surrender allowances equal to their actual emissions. Launched in 2005; current price (2024): approximately 60-70 EUR/tCO₂.

Example: A German cement plant covered by EU ETS at a carbon price of 65 EUR/tonne faces an annual cost of approximately 55 million EUR for its 850,000 tCO₂ emissions — making CCS economically attractive.

See also: Carbon Pricing, CBAM, CCS

F

Feedstock

The raw input material used in a production process. In the context of biochar, feedstock refers to the biomass material (crop residue, wood waste, manure, etc.) that is pyrolysed to produce biochar. Feedstock selection affects biochar quality, yield, stability (H:Corg ratio), and carbon accounting under VM0042. Feedstocks must be waste materials (not purpose-grown energy crops) for VM0042 eligibility.

Example: Rice straw, wheat straw, coconut shell, sugarcane bagasse, bamboo prunings.

See also: Biochar, Pyrolysis, VM0042

Flue Gas

The gas mixture produced by combustion in a boiler, furnace, or kiln that exits through a chimney or stack. In coal power plants, flue gas typically contains: 10–15% CO₂, 75–80% N₂, 5–10% water vapour, and trace pollutants (NO_x, SO₂, particulates). The low CO₂ concentration in flue gas (compared to pure CO₂) makes post-combustion capture more energy-intensive than capture from high-concentration sources.

Example: Cement kiln flue gas contains 14–30% CO₂ — higher than coal power plant flue gas — because limestone calcination produces concentrated CO₂.

See also: Post-Combustion Capture, Amine Scrubbing

FPO (Farmer Producer Organisation)

A type of producer company in India, registered under the Companies Act, 2013, that aggregates smallholder farmers to collectively market produce, access inputs, and undertake commercial activities. FPOs are the primary legal structure through which smallholder carbon projects are developed in India, as they provide a legal entity to hold registry accounts, sign offtake agreements, and distribute carbon revenue to members.

Example: An FPO in Madhya Pradesh with 500 wheat-farmer members develops a soil carbon project, holds the Verra registry account, and distributes carbon revenue to members at each verification event.

See also: Aggregation, Carbon Credits, Benefit Sharing

G

Geological Storage

The permanent underground storage of captured CO₂ in porous rock formations at depths typically greater than 800 metres. At these depths, CO₂ is in a supercritical state (denser than gas, less dense than liquid) and is trapped by one or more geological trapping mechanisms: structural (caprock), residual, solubility, and mineral trapping. Properly selected and monitored sites can store CO₂ for thousands to millions of years.

Example: The Utsira sandstone formation at Sleipner has safely stored over 25 million tonnes of CO₂ since 1996 with zero detected leakage.

See also: CCS, Trapping Mechanisms, Caprock, Saline Aquifer

GHG (Greenhouse Gas)

A gas that absorbs and re-emits infrared radiation, contributing to the greenhouse effect and climate warming. The main anthropogenic GHGs are: carbon dioxide (CO₂, GWP=1), methane (CH₄, GWP=84 over 20 years), nitrous oxide (N₂O, GWP=273 over 100 years), and various fluorinated gases. GHG concentrations are reported in CO₂ equivalent (CO₂e), weighting each gas by its GWP.

Example: Livestock produce methane (GWP84) — 1 tonne of CH₄ is equivalent to 84 tonnes of CO₂ in terms of 20-year warming impact.

See also: GWP, Carbon Budget, Net-Zero

Gold Standard

A non-governmental voluntary carbon standard founded by WWF and other environmental NGOs in 2003. Known for rigorous requirements on sustainable development co-benefits and community engagement, in addition to carbon accounting. Gold Standard projects typically command price premiums in voluntary markets due to their high quality and co-benefit requirements.

Example: A WASH (water, sanitation, hygiene) project in rural Bihar certified under Gold Standard for its emission reductions from efficient cookstoves and its health and gender co-benefits.

See also: VCS, Carbon Credits, Co-Benefits

Green Hydrogen

Hydrogen produced by electrolysis of water using electricity from renewable sources (solar, wind, hydro). When the electricity is from 100% renewables, green hydrogen production generates no direct CO₂ emissions. Currently more expensive than blue hydrogen but costs are falling rapidly as electrolyser technology improves and renewable energy prices decline.

Example: India's National Green Hydrogen Mission targets 5 million tonnes of green hydrogen per year by 2030.

See also: Blue Hydrogen, Electrolyser, Hydrogen

GWP (Global Warming Potential)

A measure of the relative warming effect of a greenhouse gas compared to CO₂ over a specified time horizon (typically 100 years, sometimes 20 years). A gas with GWP₁₀₀=25 causes 25 times as much warming as the same mass of CO₂ over 100 years. GWP is used to convert non-CO₂ GHG emissions into CO₂ equivalent (CO₂e) for unified reporting.

Example: Methane GWP₁₀₀=28; nitrous oxide GWP₁₀₀=273; SF₆ GWP₁₀₀=23,500.

See also: GHG, Carbon Credits, Carbon Accounting

H

Hard-to-Abate Sectors

Industries where reducing CO₂ emissions through renewable energy or electrification alone is technically or economically very challenging. Primary hard-to-abate sectors include: cement (process emissions from calcination), steel (blast furnace reduction), chemicals (feedstock emissions), aviation, shipping, and long-distance heavy transport. CCS is considered essential for decarbonising these sectors.

Example: Cement's calcination reaction releases CO₂ as a chemical byproduct — this cannot be avoided by switching to renewable energy. CCS is the only currently available solution.

See also: CCS, CBAM, Cement, Steel

H:Corg Ratio (Hydrogen to Organic Carbon)

The molar ratio of hydrogen to organic carbon in biochar. Used under the Verra VM0042 methodology as a proxy for biochar stability and permanence. A lower H:Corg ratio indicates higher aromaticity and greater resistance to decomposition. Biochar with H:Corg of 0.4 or lower is assigned the highest permanence (Class A, 100% crediting); H:Corg of 0.6–0.7 is Class C (70% crediting); above 0.7 is ineligible.

Example: Biochar produced at 550C from rice straw typically has an H:Corg of approximately 0.45–0.55 (Class B), qualifying for 85% credit rate.

See also: Biochar, VM0042, Permanence, Pyrolysis

Hydrogen Economy

An economic system that uses hydrogen as a primary energy carrier — for power generation, industrial heat, transport fuel, and energy storage. Hydrogen can be produced cleanly (green hydrogen from electrolysis, blue hydrogen from CCS+SMR) and burned or used in fuel cells without direct CO₂ emissions. The hydrogen economy is central to decarbonisation strategies in heavy industry, shipping, and aviation.

Example: Japan's government has committed to 3 million tonnes per year of hydrogen use by 2030 as part of its net-zero strategy.

See also: Green Hydrogen, Blue Hydrogen, Fuel Cell

I

Indian Carbon Market (ICM)

India's domestic mandatory carbon market, established under the Energy Conservation (Amendment) Act 2022, administered by the Bureau of Energy Efficiency (BEE). The ICM will initially cover large industrial energy users, requiring them to either reduce emissions or purchase carbon certificates. The ICM is expected to be operational by 2025–2026 and will create domestic demand for Indian carbon credits.

Example: A large Indian cement plant covered by ICM obligations can either install CCS, improve energy efficiency, or purchase carbon certificates from eligible projects.

See also: Carbon Market, EU ETS, Carbon Pricing

Injectivity

The rate at which CO₂ can be injected into a geological storage formation without exceeding safe pressure limits. Injectivity depends on formation permeability and thickness, and determines the maximum number of injection wells required and the practical injection capacity of a storage site. Low injectivity can significantly increase CCS costs.

Example: A storage site with 500 mD permeability can typically accommodate injection at rates of 1–3 Mt/year per well.

See also: Geological Storage, Permeability, CCS

ITMO (Internationally Transferred Mitigation Outcome)

The unit of currency in international carbon trading under Article 6.2 of the Paris Agreement. An ITMO is typically equivalent to one tonne of CO₂ equivalent reduced, avoided, or removed. When a country sells ITMOs to another country, the selling country must make a corresponding adjustment — removing the ITMO from its own NDC accounting — to prevent double-counting.

Example: Japan pays India for 500,000 ITMOs from renewable energy projects. Japan counts 500,000 tCO₂e toward its NDC; India subtracts 500,000 tCO₂e from its NDC accounting.

See also: Article 6, NDC, Corresponding Adjustment

IPCC (Intergovernmental Panel on Climate Change)

The United Nations body that assesses climate change science, established in 1988 by the World Meteorological Organization and UN Environment Programme. The IPCC synthesises peer-reviewed research and publishes Assessment Reports every 6–7 years. The Sixth Assessment Report (AR6), released 2021–2023, is the current scientific consensus on climate change.

Example: The IPCC AR6 states that human influence has unequivocally warmed the climate and that limiting warming to 1.5C requires rapid, deep emission cuts in all sectors.

See also: Carbon Budget, Net-Zero, CCS

L

Labile Carbon

Soil organic carbon that decomposes relatively quickly — within weeks to years — releasing CO₂ back to the atmosphere. Labile carbon includes fresh plant residues, microbial cells, and simple organic molecules. While labile carbon is important for soil biology and short-term nutrient cycling, it does not provide long-term carbon storage. Carbon project accounting focuses on building stable (not labile) soil carbon.

Example: Fresh straw incorporated into soil is labile — it will decompose in one to two growing seasons, releasing CO₂. Biochar applied to the same soil is stable — it will persist for centuries.

See also: Stable Carbon, Soil Carbon, Carbon Accounting

Leakage (Carbon Project)

An unintended increase in emissions outside the project boundary that occurs as a result of the project activity. Carbon project accounting must identify and quantify leakage. For example, if a biochar project uses crop residue that was previously sold as animal feed, the loss of that feed must be offset (livestock may substitute other higher-emission feed). Leakage deductions reduce the net credit claim.

Example: A soil carbon project stops open burning of crop residue; if that residue was previously used for cooking fuel, households may switch to kerosene — a leakage source that must be accounted for.

See also: Carbon Accounting, Additionality, MRV

LCA (Life Cycle Assessment)

A methodology for evaluating the total environmental impact of a product, process, or service across its full life cycle — from raw material extraction through production, use, and end-of-life. In CCS, LCA is used to evaluate the net climate impact of different production pathways (e.g. blue vs. green hydrogen), accounting for upstream emissions like methane from natural gas supply.

Example: An LCA of blue hydrogen must account for methane leakage from natural gas extraction and pipeline transport, not just the SMR and capture process.

See also: Blue Hydrogen, Carbon Accounting

M

Methane (CH₄)

A greenhouse gas approximately 84 times more potent than CO₂ over a 20-year period (GWP₂₀=84; GWP₁₀₀=28). Primary anthropogenic sources: livestock (enteric fermentation), rice paddies, coal mines, oil and gas operations, and landfills. Methane has a shorter atmospheric lifetime (~12 years) than CO₂, meaning rapid reductions in methane emissions have near-term climate benefits.

Example: India is one of the world's largest sources of agricultural methane from its large cattle population and extensive rice cultivation.

See also: GHG, GWP, Nitrous Oxide

Methodology (Carbon Standard)

A document published by a carbon standard body (Verra, Gold Standard, etc.) that provides the technical rules for quantifying, monitoring, and verifying emission reductions or removals for a specific project type. Projects must use an approved methodology and follow its requirements precisely. Key methodologies for this course: VM0042 (biochar), VM0017 (soil carbon), VM0047 (cookstoves), and Gold Standard soil carbon framework.

Example: A biochar project in Madhya Pradesh must follow VM0042 exactly — using its equations, sampling requirements, H:Corg thresholds, and baseline assessment protocols.

See also: VCS, Gold Standard, Carbon Credits, MRV

Mineral Trapping

The most secure and permanent geological CO₂ trapping mechanism. Over thousands to tens of thousands of years, dissolved CO₂ in brine reacts with rock-forming minerals (silicates, carbonates) to form solid carbonate minerals (calcite, dolomite, siderite). Once mineralised, CO₂ is essentially permanently stored — it has become part of the rock.

Example: In the Hellisheidi basalt formation in Iceland (Carbfix project), CO₂ has been found to mineralise within 2 years — far faster than previously thought.

See also: Geological Storage, Trapping Mechanisms, Saline Aquifer

MRV (Monitoring, Reporting, Verification)

The framework for tracking, documenting, and independently confirming the emission reductions or removals from a carbon project. Monitoring: ongoing data collection per the approved monitoring plan. Reporting: preparing the monitoring report in the format required by the carbon standard. Verification: independent third-party audit by an accredited VVB (Validation and Verification Body) confirming the accuracy of the monitoring data and calculations.

Example: After Year 2 of a biochar project, the implementing organisation submits a monitoring report to Verra; an accredited VVB conducts a site visit and data audit, and issues a verification opinion; Verra then issues VCUs based on the verified reduction figure.

See also: Carbon Credits, VVB, DMRV

N

NDC (Nationally Determined Contribution)

A country's climate action plan submitted under the Paris Agreement, specifying its emission reduction targets, policies, adaptation plans, and finance needs. NDCs are updated every 5 years and are expected to become progressively more ambitious. India's current NDC targets include achieving 50% non-fossil electricity capacity by 2030 and reducing the emissions intensity of GDP by 45% from 2005 levels.

Example: India's NDC commits to 45% reduction in GDP emissions intensity by 2030 compared to 2005 — a target it is broadly on track to meet.

See also: Paris Agreement, Article 6, ITMO

Net-Zero

A state in which the anthropogenic greenhouse gas emissions entering the atmosphere are balanced by anthropogenic removals — so that net additions to atmospheric GHG concentration are zero. Net-zero does not mean zero emissions; it means residual emissions from hard-to-abate sectors are counterbalanced by CDR. Most countries have 2050 net-zero targets; India has committed to 2070.

Example: A net-zero steel plant might emit 100 kt of CO₂ per year that cannot yet be avoided, but sequesters 100 kt/year via soil carbon and biochar projects, achieving net zero.

See also: CDR, Carbon Budget, Hard-to-Abate

Nitrous Oxide (N₂O)

A powerful greenhouse gas with GWP₁₀₀=273 (273 times more potent than CO₂ over 100 years). Primary agricultural source: nitrogen fertiliser application, which promotes microbial processes in soil that produce and release N₂O. Reducing N₂O from agriculture — through precision fertiliser application, nitrification inhibitors, and optimal timing — is an important co-benefit of regenerative agriculture practices.

Example: 1 kg of N₂O released from soil has the same 100-year warming impact as 273 kg of CO₂.

See also: GHG, GWP, Regenerative Agriculture

O

Offtake Agreement

A binding contract between a carbon project developer (seller) and a carbon credit buyer that specifies the volume, price, standard, vintage, and delivery terms for carbon credits over a fixed period. Offtake agreements reduce revenue risk for project developers (enabling project finance) and secure supply for buyers with net-zero commitments. Key terms include: minimum delivery volume, floor price, force majeure provisions, and replacement obligations in case of reversal.

Example: A corporate buyer commits to purchasing 10,000 tCO₂e/year at a floor price of USD 28/tonne for 5 years from a Gold Standard soil carbon project.

See also: Carbon Credits, Carbon Market, Project Finance

Open Burning

The practice of burning organic materials — crop residue, municipal waste, forest clearing debris — in the open air without any emission controls. Open burning releases CO₂, methane, nitrous oxide, black carbon, and toxic pollutants directly to the atmosphere. In India, open burning of 65+ million tonnes of crop residue annually is a major source of air pollution and GHG emissions, and is the primary target of smallholder biochar projects.

Example: Punjab alone burns approximately 15–20 million tonnes of rice straw each year, causing severe air pollution across North India and releasing approximately 10 Mt CO₂e.

See also: Biochar, Crop Residue, Avoided Emissions

Oxyfuel Combustion

A combustion technology in which fuel is burned in a mixture of pure oxygen (from an air separation unit) and recirculated flue gas, rather than air. Because nitrogen is absent, the combustion products are predominantly CO₂ and water vapour. After condensing the water, a highly concentrated CO₂ stream is obtained, which can be directly compressed for storage — without needing a separate chemical absorption step.

Example: An oxyfuel cement kiln produces a flue gas of 80%+ CO₂, which can be compressed and stored with minimal additional processing.

See also: CCS, Post-Combustion Capture, Pre-Combustion Capture

P

Paris Agreement

The landmark international climate treaty adopted at COP21 in Paris in December 2015, entered into force November 2016. 196 parties have signed. The agreement's central goals: (1) limit global average temperature increase to well below 2C above pre-industrial levels, and pursue efforts to limit to 1.5C; (2) increase countries' ability to adapt to climate change; (3) make finance flows consistent with low-emission development. Countries submit NDCs every 5 years.

Example: The Paris Agreement's 1.5C temperature goal has become the organising objective of the global climate policy and carbon market ecosystem.

See also: NDC, Carbon Budget, Article 6

Permanence

The duration for which carbon is stored out of the atmosphere. A key criterion for carbon credit quality. Geological storage provides very high permanence (thousands to millions of years). Biochar provides high permanence (hundreds to thousands of years; H:Corg-dependent). Soil carbon provides medium permanence (years to decades; reversible by tillage or drought). Forest-based credits provide lower permanence (risk of fire, disease, logging).

Example: Under VM0042, biochar with H:Corg of 0.4 receives 100% permanence credit; with H:Corg of 0.6, receives 85%.

See also: Carbon Credits, Buffer Pool, Geological Storage

Post-Combustion Capture

Carbon capture technology applied after fuel combustion, removing CO₂ from the exhaust gas (flue gas). The most widely deployed CCS approach, applicable as a retrofit to existing facilities. Post-combustion capture using amine solvents is commercially mature and has been demonstrated at scale at Boundary Dam (Canada) and in natural gas processing worldwide.

Example: Boundary Dam: flue gas at 12% CO₂ contacts Cansolv amine solvent, capturing 90%+ of CO₂ before the cleaned gas exits the chimney.

See also: Amine Scrubbing, Pre-Combustion Capture, CCS

Pre-Combustion Capture

Carbon capture technology applied before combustion, capturing CO₂ from a fuel stream that has been converted to syngas via reforming or gasification. Pre-combustion capture is most applicable to hydrogen production (SMR), IGCC power plants, and natural gas processing. The CO₂ concentration in pre-combustion streams (15–40%) is higher than in post-combustion flue gas (~12–15%), making separation cheaper per tonne.

Example: Quest (Canada): CO₂ is removed from the SMR process stream before the hydrogen is burned, using the Benfield potassium carbonate process.

See also: Syngas, Hydrogen, SMR, CCS

Pyrolysis

The thermal decomposition of organic material at elevated temperatures (300–700C) in the absence of oxygen. Pyrolysis of biomass produces three products: biochar (a solid carbon-rich residue, 25–35% of dry input weight), bio-oil (a liquid mixture of organic compounds, 25–40%), and syngas (a mixture of CO, H₂, CH₄, CO₂, 15–25%). The temperature and residence time of pyrolysis determine the proportion and quality of each product.

Example: Slow pyrolysis at 450C for 60 minutes yields approximately 30% biochar with H:Corg of ~0.5 (Class B under VM0042).

See also: Biochar, TLUD, Carbon Sequestration

Regenerative Agriculture

A set of farming practices that aims to restore and enhance soil health, ecosystem function, and biodiversity while maintaining productive agriculture. Key practices include: no-till or minimum tillage, diverse crop rotations, cover cropping, compost and organic matter addition, integrated pest management, and agroforestry. Regenerative agriculture sequesters carbon in soil, reduces N₂O emissions, and improves water infiltration and drought resilience.

Example: A 500-hectare farm in Maharashtra transitioning from monoculture cotton to a diverse rotation with cover crops and reduced tillage can sequester 0.5–1.5 tCO₂e/ha/year in soil organic carbon.

See also: Soil Carbon, Agroforestry, CDR

Residual Trapping

A geological CO₂ trapping mechanism in which CO₂ droplets or ganglia are physically trapped in rock pore spaces by capillary pressure forces — much like water trapped in a sponge. As the CO₂ plume migrates through a porous formation, some CO₂ is left behind as immobile residual saturation. This mechanism begins to be significant after approximately 100 years and becomes dominant between 100 and 1,000 years of storage.

Example: At Sleipner, 4D seismic analysis shows that portions of the CO₂ plume that migrated upward in early years are now immobilised as residual saturation, contributing to long-term security.

See also: Geological Storage, Trapping Mechanisms, Saline Aquifer

Reversal (Carbon)

The re-release of carbon that was previously sequestered and credited as a carbon removal. Reversals can be intentional (land use change, practice abandonment) or unintentional (forest fire, drought-induced SOC loss). Carbon standards manage reversal risk through buffer pools. Geological storage is considered permanent and not subject to reversal under standard frameworks.

Example: A forest carbon project's trees are destroyed by wildfire — the sequestered carbon is re-emitted, creating a reversal. The project's buffer pool covers the reversal without requiring buyers to return credits.

See also: Permanence, Buffer Pool, Carbon Credits

S

Saline Aquifer

An underground layer of porous rock saturated with salty water (typically >10,000 mg/L total dissolved solids) that is unsuitable for drinking or irrigation. Deep saline aquifers at depths of 800+ metres are the most abundant geological CO₂ storage resource globally, with estimated total capacity of 100,000+ GtCO₂ — far exceeding any realistic CCS deployment scale.

Example: The Utsira sandstone (Sleipner), Johansen sandstone (Northern Lights), and Basal Cambrian Sands (Quest) are all deep saline aquifers used for CO₂ storage.

See also: Geological Storage, Trapping Mechanisms, CCS

SMR (Steam Methane Reforming)

The most common industrial process for producing hydrogen from natural gas. Natural gas (methane, CH₄) reacts with steam at ~850°C over a nickel catalyst, producing syngas (CO + H₂). A subsequent water-gas shift reaction converts CO + H₂O to CO₂ + H₂, yielding a hydrogen-rich stream and concentrated CO₂. With CCS (blue hydrogen), the CO₂ is captured and stored; without CCS, it is vented (grey hydrogen).

Example: India produces approximately 6 million tonnes of hydrogen annually, almost entirely via SMR without CCS (grey hydrogen).

See also: Hydrogen, Blue Hydrogen, Pre-Combustion Capture

SOC (Soil Organic Carbon)

The carbon component of soil organic matter — the total stock of carbon held in living and dead organic material in the soil. SOC is measured as a percentage of soil mass or as tonnes per hectare to a specified depth. SOC is the primary metric for soil carbon sequestration projects. Healthy agricultural soils typically have 1–3% SOC in the top 30 cm; degraded soils may have 0.3–0.8%.

Example: A field with 1.2% SOC in the top 30 cm, bulk density of 1.3 t/m³, contains approximately 47 tC/ha = 172 tCO₂e/ha.

See also: Soil Carbon, Carbon Accounting, MRV

Soil Carbon

Carbon stored in soil organic matter — including plant debris, microbial biomass, humus, and charcoal residues. Soil is the largest terrestrial carbon pool (approximately 2,500 GtC globally — more than the atmosphere and all above-ground vegetation combined). Agricultural soils have lost 50–70% of their original carbon stocks through tillage, erosion, and loss of organic matter return. Restoring soil carbon is a significant CDR opportunity.

Example: India's agricultural soils contain approximately 50–150 tC/ha; improving management could add 0.5–1.5 tCO₂e/ha/year.

See also: SOC, Regenerative Agriculture, CDR, MRV

Solubility Trapping

A geological CO₂ trapping mechanism in which CO₂ dissolves into the brine surrounding it in the storage formation. CO₂-saturated brine is denser than pure brine, so it sinks — removing CO₂ from the potentially mobile free-phase plume and placing it in a more stable dissolved form. Solubility trapping becomes significant between 1,000 and 10,000 years of storage.

Example: In the Utsira formation at Sleipner, dissolution of CO₂ into brine has begun to increase and will dominate trapping over the next several thousand years.

See also: Trapping Mechanisms, Geological Storage, Mineral Trapping

Structural Trapping

The primary and most immediate geological CO₂ trapping mechanism. Injected CO₂ is less dense than surrounding brine and migrates upward until it encounters an impermeable caprock — a seal formation above the porous reservoir. The CO₂ accumulates under the caprock in a structural trap (an anticlinal dome or fault-sealed structure). Structural trapping begins immediately upon injection and provides the primary containment in the first 0–100 years.

Example: At Quest, the Basal Cambrian Sands are sealed by the overlying Proterozoic shale formations — structural trapping begins the moment CO₂ is injected.

See also: Caprock, Geological Storage, Trapping Mechanisms

Supercritical CO₂

CO₂ in a physical state above its critical temperature (31.1°C) and critical pressure (73.8 bar), at which it exhibits properties of both a liquid and a gas. In supercritical state, CO₂ is denser than gas (more efficient to store) but less dense than water or brine. At depths greater than approximately 800 metres, reservoir temperature and pressure typically maintain CO₂ in supercritical state, which is the optimal condition for geological storage.

Example: At 2,600 m depth (Northern Lights Johansen formation), temperature (~70°C) and pressure (~260 bar) keep injected CO₂ in supercritical state.

See also: Geological Storage, CCS

TLUD (Top-Lit Updraft)

A simple and efficient biomass gasifier-kiln design commonly used for smallholder-scale biochar production. In a TLUD, biomass is loaded into a cylindrical vessel, ignited from the top, and air enters from the bottom through small holes. Combustion progresses downward through the biomass; volatile gases are burned in the upper flame zone. TLUD kilns produce minimal smoke (gases are combusted internally), making them more suitable for use near dwellings than traditional earthen kilns.

Example: A 200-litre steel drum TLUD kiln costs approximately INR 2,000–3,000 to fabricate, processes 25–30 kg of dry crop residue per batch, and produces 6–8 kg of biochar in 60–90 minutes.

See also: Biochar, Pyrolysis, Feedstock

Trapping Mechanisms

The geological processes that secure CO₂ injected into storage formations. Four main mechanisms act sequentially: (1) Structural trapping — CO₂ physically held under impermeable caprock; (2) Residual trapping — CO₂ droplets immobilised in pore spaces by capillary forces; (3) Solubility trapping — CO₂ dissolved into brine; (4) Mineral trapping — CO₂ reacted with rock minerals to form solid carbonates. Together, these mechanisms provide progressively more secure containment over time.

Example: Sleipner monitoring data shows all four trapping mechanisms in evidence, with residual and solubility trapping increasing in significance over 28 years.

See also: Geological Storage, CCS, Permanence

U

UNFCCC

United Nations Framework Convention on Climate Change — the foundational international climate treaty adopted at the Rio Earth Summit in 1992. The UNFCCC established the framework for international climate negotiations; subsequent agreements (Kyoto Protocol, Paris Agreement) operate under its auspices. India ratified the UNFCCC in 1993 and is an active participant in negotiations.

Example: COP — the Conference of the Parties to the UNFCCC — is the annual meeting where countries review progress and make decisions under the treaty.

See also: Paris Agreement, NDC, COP

V

Verra / VCS (Verified Carbon Standard)

The world's largest voluntary carbon market standard and registry organisation, based in Washington DC. Verra operates the Verified Carbon Standard (VCS), under which carbon projects are developed, verified, and issued Verified Carbon Units (VCUs). Over 1,800 projects have been certified under VCS, generating over 1 billion VCUs to date. Key methodologies for this course: VM0042 (biochar) and VM0017 (soil carbon).

Example: A soil carbon project in Telangana applies for VCS certification under VM0017, preparing a PDD, undergoing validation, and receiving VCUs after first verification.

See also: VM0042, Gold Standard, Carbon Credits

VCU (Verified Carbon Unit)

The tradeable carbon credit issued by Verra under the Verified Carbon Standard. One VCU represents one tonne of CO₂ equivalent reduced, avoided, or removed from the atmosphere, and verified by an accredited VVB. VCUs can be retired by buyers to offset emissions or held for resale in the voluntary carbon market.

Example: A corporate buyer retires 5,000 VCUs from a VM0042 biochar project to offset its Scope 1 emissions for a financial year.

See also: Verra, Carbon Credits, Retirement

Voluntary Carbon Market (VCM)

A decentralised marketplace where organisations voluntarily buy and sell carbon credits to offset their emissions or demonstrate climate leadership — as opposed to compliance markets where participation is legally mandatory. The VCM is driven by corporate net-zero commitments, brand value, and investor/customer expectations. Market size reached approximately USD 2 billion in 2022 and is projected to grow to USD 50–100 billion by 2030–2035.

Example: A fashion brand buys 10,000 Gold Standard VERs from an agroforestry project in Jharkhand to offset the emissions of a product launch event.

See also: Carbon Market, Carbon Credits, Gold Standard

VVB (Validation and Verification Body)

An accredited, independent third-party organisation that validates carbon project design documents (confirming methodology compliance and reasonable baseline) and verifies monitoring reports (confirming that claimed emission reductions actually occurred). VVBs must be accredited by the relevant carbon standard (Verra, Gold Standard, etc.) and demonstrate expertise in the relevant sector and methodology.

Example: DNV (Det Norske Veritas), Bureau Veritas, SGS, and South Pole are among the major accredited VVBs active in Indian carbon projects.

See also: MRV, Carbon Credits, Methodology

W

Water-Gas Shift Reaction

A chemical reaction used in industrial hydrogen production: $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$. The reaction converts carbon monoxide (CO) in syngas to CO₂, simultaneously producing additional hydrogen. The resulting high-concentration CO₂ stream (15–40%) is far cheaper to separate than dilute flue gas CO₂, making pre-combustion capture (as at the Quest project) more energy-efficient than post-combustion capture.

Example: At Quest's Scotford SMR plants, the water-gas shift reaction produces a gas stream of approximately 20% CO₂ — which the Benfield potassium carbonate system removes at substantially lower energy cost than amine scrubbing of dilute flue gas.

See also: SMR, Syngas, Pre-Combustion Capture, Hydrogen
